

International Islamic University Chittagong
Center for General Education
Final Exam- Autumn 2023

Course Code: URED- 3503 (URED- 3101 for LLB)
Course Title: Political Thoughts and Social Behavior
Time: 2.5 Hours **Full Marks: 50**

[Answer the followings. All questions are of equal value]

✓ 1. Discuss about the importance and rules of marriage as the basis of Islamic family.

2. "The rights necessarily are followed by some duties"- based on this statement discuss the rights and duties of women in Islam.

Or

✓ Compare the status of women in Islam with what was before Islam.

✓ 3. State some principles of Islamic economic system with their explanation that can help us to deal our economic affairs complying with *shariah*.

✓ 4. Islamic law has provided mankind with appropriate instructions and guidance regarding dress for man and woman. Elaborate those guidelines after mentioning the purposes of dress in Islam according to the *ayah* 26 of *surah al-a'raf*.

5. Write an essay on "How should we bring up our children" according to the demand of Islamic *Shariah* and the situation of contemporary life.

Or

Write shortly about the following (any five):

- ✓ a) Duty towards needy and orphan
- b) Rules of greetings in Islam
- c) Importance of tolerance
- d) Necessity toward the honesty
- e) Importance of the reliance on Allah
- f) Rules of sneezing in Islam.

International Islamic University Chittagong

Morality Development Program (MDP)

Semester End Examination: Autumn 2023

5th Semester (Other than Shariah Faculty)

Course Code: MDP-3505, Course Title: Concepts of Moral Development-II

Full Marks: 50

Time: 2.5 Hours

[Answer the following questions. All parts of a question must be answered sequentially. Figures in the right margin indicate full marks.]

1. (a) List the drawbacks of Western culture adoption, especially among the youth of today. 5

(b) Mention some characteristics of Islamic culture. Distinguish between Islamic culture and western culture in brief. 5

2. Explain the guidelines of dress code for both male and female in Islamic points of view. 10

OR

Explains the importance of family in the eyes of Islam. Also explain the reason of breaking the family. 10

3. (a) Explain conspiracy theory against Islam and Muslim. 5

(b) Mention some hadith studies on the day of Judgment. 5

4. (a) What are the basic human rights in Islam? Explain the rights of women in light of Quran and Sunnah. 10

OR

Write short notes on: Freemason, Hadith on Day of Judgment, Basis of unity of the Muslim Ummah, Colonization of Muslim countries, right to protect the property 10

5. (a) Entail Freemasonry around the globe in brief. 5

(b) Describe in brief the colonization of Islamic countries in world by European colonial powers. 5

[Answer all the questions, Figures in the right hand margin indicate full marks]

Group A

1.	a)	What is an interrupt? Describe the types of interrupts with examples.	CO3	Un	5
1.	b)	Suppose you have four input devices: Disk, Printer, Reader, and Keyboard. All these devices are sending interrupts to the Microprocessor 8086. Note that the device names are listed in descending order of priority.. How does Microprocessor 8086 handle the above situation in the daisy chain process? Explain the handling process with the proper figure.	CO3	An	5
OR					
1.	b)	What role does the 8259A Programmable Interrupt Controller (PIC) play in managing interrupt priorities and vectoring in microprocessor systems, and how does the expand-ability of the 8259A to accept up to 64 interrupt requests impact the scalability and performance of these systems?	CO3	An	5
2.	a)	What is the significance of using interrupts in the context of keyboard input, and how does the generation of a single pulse by the keyboard encode of the system? Explain with timeline diagram	CO1	Un	5
2.	b)	How does the NMI function in terms of its triggering conditions and its role in detecting critical events such as power failures and parity errors in microprocessor-based systems, and why is it essential for maintaining system reliability?	CO1	Un	5
OR					
2.	b)	What is bit slicing in microprocessor design, and how does this approach enable the construction of specialized, high-performance processors for specific applications or functions by combining multiple bit-slice units?	CO1	Un	5
Group B					
3.	a)	What are the fundamental differences between isolated I/O and memory-mapped I/O methods, and how do these two approaches affect the interaction between a microprocessor and external devices in a computer system?	CO3	Un	5
3.	b)	How can you perform Multiplication and Division operations in Microcontroller 8051? Explain the process.	CO3	Un	5
OR					
3.	b)	What are the primary components involved in basic input and output operations within a microprocessor system, and how are these operations defined in terms of IN and OUT, referring to data movement between the microprocessor and I/O devices? Explain with proper diagram.	CO3	Un	5
4.	a)	Recently, Amit and Arafat learned about the 8051 microcontroller. However, they are aware that there is a pin that separates address and data lines and another pin that disables internal ROM, but they are having trouble recognizing them. Help them by describing these pins.	CO2	Ap	5
4.	b)	What are the key functions of the main pins in the Intel 8051 microcontroller, as depicted in the pinout diagram, and how do these pins contribute to the microcontroller's operation and its ability to interface with external devices?	CO2	Ap	5
5.	a)	What is a System-on-a-Chip (SoC)? Describe each of its components with a diagram.	CO3	An	5
5.	b)	What is the significance of the PLD (Programmable Logic Device) in the context of decoding I/O addresses in embedded systems? How does the addition of eight address lines (A15-A8) affect the decoding process, and how does the PLD generate address strobes for I/O ports?	CO3	Un	5
OR					
5.	b)	Provide a brief explanation of modeling embedded system behavior, illustrated with an example of a finite state machine model.	CO3	Un	5

International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE Final Examination, Spring-2023
Course Code: CSE-3529 Course Title: System Analysis and Design
Marks: 50 Time: 2 Hours 30 Minutes

Instructions: Answer all the questions. Use separate answer script for Group A and Group B.
Figures on the right margin indicate marks.

Group: A

- | | | |
|--|---|-----|
| 1(a) "A structured repository contains the descriptions of data flow diagram of data elements" – Identify which type of structured analysis tools is mentioned in the above statement. Briefly discuss about it with its significance. | 2 | CO3 |
| 1(b) Differentiate between "Decision Tree" and "Decision Table" with examples. | 3 | CO3 |
| 1(c) We wish to draw the DFD for a distance education university. The enrolment process works as follows:
Students <u>send in an application form</u> containing their personal details and their desired course. The university checks that the <u>course is available</u> and the student has necessary academic qualifications. If the course is available the student is enrolled in the course and the <u>university confirmed</u> the enrolment by sending a <u>confirmation letter</u> to the student. If the course is unavailable the student is sent a rejection a letter.
(i) Find out and describe the suitable DFD context level input and output.
(ii) Find out the level 0 DFD | 5 | CO3 |

OR

"Amazon.com" is a globally reputed e-commerce platform that offers an extensive and diverse range of products such as clothing, electronic gadgets, household items and many more. This site has the following criterion:

- i. Users/Customers can create accounts, providing personal information, usernames, and passwords for secure access.
- ii. Sellers needs to list products, including associated media (images, descriptions), for customers to view and purchase.
- iii. Registered users, both customers and sellers, interact with the Amazon online shopping system after authentication, including browsing and shopping.

Draw a DFD (Data Flow Diagram) according to the above scenario including all necessary symbols of DFD and the following –

- A. Find all the external entities
- B. Design Context diagram or Level-0 Diagram
- C. Design Level-1 diagram using balancing and leveling techniques

2(a) State the objectives of input design. Explain the differences between good form design and bad form design with 3 different design examples. 5 CO3

2(b) A department of a company has a supervisor and at least one employee. Employees must be assigned to at least one, but possibly more departments. At least one employee is assigned to a project, but an employee may be on vacation and not assigned to any projects. The important data fields are the names of the departments, the projects, the supervisors and the employees, as well as the supervisor and employee number and a unique project number. Draw the ER diagram related to the scenario described above and provide short explanation of the diagram. 5 CO3

OR

"IIUC Computer Club" is one of the most active and promising student clubs of Department of Computer Science & Engineering at International Islamic University Chittagong. This CSE club dedicatedly works for students and organizes different events that enhances the academic curriculum. Students of CSE department by default gets its membership and participates in various activities organized by club.

Draw an ERD for the above scenario with proper symbols.

Group: B

3(a) What is system testing? Is testing compulsory for a system or not? Justify your answer with example. 5 CO3

OR

Assume you are a software engineer of a developing firm. Write the possible consequences may occur if you do not make any kind of testing before handover your software (Discuss for both user and developer point of view).

3(b) Define the following terms with suitable example: Alpha testing, Beta testing, White box testing and Black box testing. 5 CO3

4(a) State system implementation. Mention the six major activities that ensures system implementation process. 2 CO1

4(b) Write about the types of documentation of any system with example. 4 CO1

4(c) Mr. XYZ is a project manager of a mobile game's development software firm. He has a dedicated group of teams that works for app maintenance. Now if any maintenance request comes to his organization, what do you think how will the project manager handle it? Explain with figure. 4 CO2

5(a) What is threat? Briefly explain about different kinds of threats. Write some criteria for software and hardware selection. 5 CO2

5(b) What is security? Write some security issues with examples. How to prevent vulnerability from a system? 5 CO2

OR

Write short notes on— i) Encryption ii) Interruption iii) Data Vulnerability
iv) System Testing iv) Recovery Testing

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn 2023
Course Code: CSE 3521 Course Title: Computer Architecture
Total marks: 50 Time: 2 hours 30 minutes

[Answer all questions from **Group-A** and **Group-B**
 Separate answer script must be used for Group-A and Group-B.]

Group-A

- | | | |
|--|---|-----|
| 1.a) What is hardwired control? How it is different from micro-programmed control? | 2 | CO1 |
| b) Suppose you are given an instruction <code>sub \$s1,\$t3,\$t4</code> . Describe the complete operation of the datapath using the instruction with a figure. | 5 | CO1 |
| Or, | | |
| Illustrate the following instruction with three bus and two bus CPU system and explain which is better? <code>ADD R3, R2, R1</code> | | |
| c) Why a single-cycle implementation is not used? justify the answer. | 3 | CO2 |
| 2.a) Do you think that the functions of memory data register and instruction register are the same? If not then how? | 3 | CO1 |
| b) Write down the basic characteristics of RISC and CISC architectures. | 5 | CO1 |
| Or, | | |
| Design a multi-cycle data path for handling basic instruction of MIPS. | | |
| c) What changes are required for the multiplexer in multicycle datapath? | 2 | CO2 |

Group-B

- | | | |
|---|---|-----|
| 3.a) Does the use of pipelining reduce, increase or leave unchanged the execution time of a single machine instruction? | 3 | CO3 |
| b) How can you resolve data hazard -explain with figure. | 5 | CO1 |
| c) Dynamic pipelining is more complicated than the traditional or static pipelining – why? | 2 | CO2 |
| 4.a) Remark on the purpose, scope and mechanisms of the followings: i) Programmed I/O
ii) Interrupt Initiated I/O iii) Direct Memory Access. | 3 | CO4 |
| b) What is a DMA controller? Write the configuration of the DMA controller. | 5 | CO1 |
| Or | | |
| What is TLB? How can you reduce page fault? Explain logically? | | |
| c) Define miss penalty, cache miss, hit rate, tag in caching. | 2 | CO1 |

5.a) Find the Location of a memory block with address 14 in a cache with 8 blocks with different degrees of associativity

5 CO2

Or

When will the page faults occur? What is the procedure for handling the page fault?

b) What are the key factors for designing a virtual memory? What are similarities between cache and virtual memory?

3 CO1

c) Show the comparative analysis of write through scheme and write back scheme.

2 CO1

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Exam, Autumn 2023

Course Title: **Electrical Drive and Instrumentation**

Full Marks: 50

Course Code: **CSE-2421**

Time: 2 hours 30 minutes

<u>Group A</u>					
1.	a)	What is slip? Explain the production of rotational magnetic field in induction motor with appropriate diagram.	CO1	Rm	3
1	b)	Draw the transformer equivalent circuit of 3-Phase Induction Motor.	CO1	Un	3
1.	c)	A 208V, 10-hp, four-pole, 50-Hz, Y-connected induction motor has a full-load slip of 5%. (i) What is the synchronous speed of this motor? (ii) What is the rotor speed of this motor at the rated load? (iii) What is the rotor frequency of this motor at the rated load? (iv) What is the shaft torque of this motor at the rated load?	CO1	An	4
2.	a)	Discuss the full step mode operation of the Stepper Motor with neat diagrams.	CO1	Un	5
2.	b)	Explain the working principle of Variable frequency drive with proper diagram. OR	CO3	Un	5
2.	b)	What is the difference between Induction Motor and Synchronous Motor? Explain a method to make Synchronous Motor Self-Starting.	CO3	Un	5
<u>Group B</u>					
3.	a)	What are the different types of errors that can occur in the process of measurement? Briefly explain systematic errors.	CO3	An	5
3.	b)	What do you understand by the term measurement and instrumentation. What steps comprise up the Generalized Measuring System? Also, provide an example for each step.	CO3	Ap	5
4.	a)	Explain the operation of Solar Cell with proper diagram.	CO4	Ap	5
4.	b)	Derive the output voltage equation of logarithmic amplifier with transistor and op-amp. OR	CO4	Ap	5
4.	b)	What is Optoelectronics. Mention different types of optoelectronic devices and explain the basic principle of Light Emitting Diode (LED).	CO4	Un	5
5.	a)	Sketch 7-segment display with truth table for both common cathode mode and common anode mode.	CO4	Un	5
5.	b)	What is transducer? Prove the gauge factor equation for strain gauge.	CO4	Un	5

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Term Examination, Autumn-2023

Course Code: CSE-3527 Course Title: Compiler

Total marks: 50 Time: 2.5 Hours

[Answer all the questions, Figures in the right hand margin indicate full marks]

CO

Group A

1. a) Construct CLR(1) parsing table for the following grammar: A 4 CO2
 $S \rightarrow Aa \mid bAc \mid dc \mid bda$ P
 $A \rightarrow a$
- b) If the following grammar is suitable for LL (1) parsing then check whether the string $qqqqr\$$ can be generated from the grammar by showing step by step operations. A 6 CO3
P
 $A \rightarrow Q$
 $Q \rightarrow St \mid Rr$
 $S \rightarrow sS \mid \epsilon$
 $R \rightarrow qR \mid \epsilon$
2. a) Consider the following grammar: A 6 CO3
P
 $X \rightarrow X + Y / Y$
 $Y \rightarrow Y d F / F$
 $F \rightarrow (X) / id$
(i) Find the first and follow functions for the given CFG with Left Recursive production rules.
(ii) Is the above grammar is LL(1) grammar? Answer the question with proper justification.
- b) Consider the following grammar: A 4 CO3
P
 $E \rightarrow T E'$
 $E' \rightarrow + T E' \mid \square$
 $T \rightarrow F T'$
 $T' \rightarrow * F T' \mid \square$
 $F \rightarrow (E) \mid id$
Show the sequence of moves on input $id*id+id$ using the nonrecursive predictive parser algorithm.
- OR,
2. a) Consider the grammar: A 4 CO3
P
 $S \rightarrow (L) \mid a$
 $L \rightarrow L, S \mid S$
1. Construct a right most derivation for $(a, (a, a))$ parse tree.
2. Show the steps of shift reduce parser considering right most derivation for $(a, (a, a))$
- b) Consider the following grammar: A 6 CO3
P
 $S \rightarrow MM$
 $M \rightarrow mM \mid n$
1. Construct the simple LR parsing table by using canonical items
2. Explain the parsing strategies with stack implementation for input string "mmnn"

Group B

3. a) Define the Synthesized attributes and inherited attributes. Using the following production rules and semantic rules draw the annotated parse tree and dependency graph for the sentence float id1, id2, id3.

Ap 1+ CO3
5

PRODUCTION	SEMANTIC RULES
1) $D \rightarrow T L$	$L.inh = T.type$
2) $T \rightarrow \text{int}$	$T.type = \text{integer}$
3) $T \rightarrow \text{float}$	$T.type = \text{float}$
4) $L \rightarrow L_1, \text{id}$	$L_1.inh = L.inh$ $addType(id.entry, L.inh)$
5) $L \rightarrow \text{id}$	$addType(id.entry, L.inh)$

- b) Mention the differences among parse tree, syntax tree and DAG. U 2 CO2
- c) Write the algorithm for Left factoring. R 2 CO3
4. a) Explain two strategies for dynamic storage allocation. R 4 CO1
- b) By using SDD of simple desk calculator, construct annotated parse trees for the following expression: $3*5 + 4n$ U 4 CO1
- c) Define DAG. Draw the DAG for this basic block: U 2 CO3
- $a = b + c$
 $b = b - d$
 $c = c + d$
 $e = b + c$
5. a) Consider the following C code and answer the questions (i) to (iii) Ap 5 CO3
- ```

Void production(int a[], int b[]){
 int prod=0, i=1;
 do{ prod=prod + a[i]*b[i];
 i=i+1;
 }while(i<=20);
 printf("product=%d", prod);
}

```
- i) Translate the C code into 3 address code
- ii) Identify the basic block in 3 address code
- iii) Construct the flow graph from the 3-address code
- b) Define basic block and flow graph with example. U 3 CO1
- c) Fill in the blanks: R 2 CO1
- i) Data type grammar use \_\_\_\_ attributes
- ii) Symbol table is generated in \_\_\_\_ phase.
- iii) Operator precedence parser is \_\_\_\_ parser
- iv) \_\_\_\_ parser can parse using ambiguous grammar.
- OR,
5. a) Translate the following expression to quadruple, triple and indirect triple- Ap 5 CO3
- $a + b * c / e + b * c$
- b) What do you mean by Peephole optimization? Write down the U 3 CO1
- characteristics of Peephole optimization with example.
- c) Explain the primary tasks of code generator. R 2 CO1